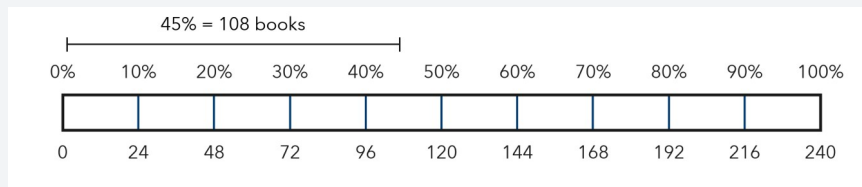


LESSON 4-4 • SESSION 1 • EXPLAIN YOUR THINKING**Find the Percent of a Number**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS *part* (parte) · *whole* (entero) · *equation* (ecuación)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

A shelf has 240 books. 45% of them are mysteries. Use the percent bar and an equation to find how many books are mysteries.

**HOW I SOLVED IT**

- 1 Find 10%.** $240 \div 10 = 24$ books.
- 2 Build 45%.** $40\% = 4 \times 24 = 96$. $5\% = \text{half of } 24 = 12$. $96 + 12 = 108$ books.
- 3 Check with an equation.** $0.45 \times 240 = 108$. 108 is less than half of 240 (120), so it makes sense.

I SAY

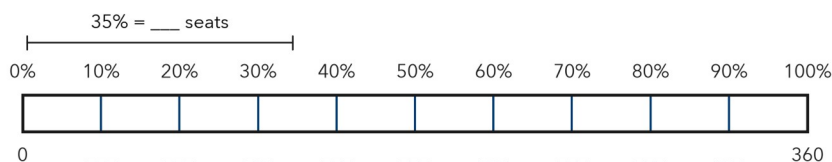
"The whole is **240**. 10% of it is **24**, so 45% is **108**."
 "Check: **$0.45 \times 240 = 108$** ."

I WRITE

I know 45% of 240 is **108** because **10% is 24, so 40% is 96, and 5% is 12; $96 + 12 = 108$** .

2 • DRAW IT*your turn: label the whole, the part and the percent*

A theater has 360 seats. 35% of the seats are in the balcony. Use the percent bar and an equation to find how many seats are in the balcony.

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Find 10% of 360: $360 \div 10 = \underline{\hspace{2cm}}$.
- 2** Build 35%: $30\% = \underline{\hspace{2cm}}$ and $5\% = \underline{\hspace{2cm}}$. Add them.
- 3** Check: $0.35 \times 360 = \underline{\hspace{2cm}}$.

3 • SAY IT*tell a partner*

Partner A: "The whole is . 10% of it is , so 35% is ." Partner B: "Check with an equation: $\times$ = . Is the part less than the whole?"

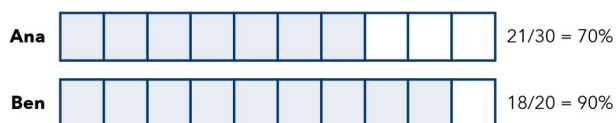
4 • WRITE IT*pick a starter · name the whole · use units***WORD BANK** 10% of · half of · the whole · the part · equation · less than half**Because** I know 35% of 360 is because .**But** Some students multiply 35×360 , but .**So** 35% is less than half of the whole, so .

LESSON 4-4 • SESSION 2 • EXPLAIN YOUR THINKING**Compare Percents**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS **part-to-whole ratio** (*razón de parte al entero*) · **common whole** (*entero común*) · **compare** (*comparar*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

Ana got 21 of 30 quiz points. Ben got 18 of 20. Ana got more points. Who did better on the quiz?

**HOW I SOLVED IT**

- 1 Ana.** $21/30 = 7/10 = 70/100 = 70\%$. Shade 7 of 10 parts.
- 2 Ben.** $18/20 = 9/10 = 90/100 = 90\%$. Shade 9 of 10 parts.
- 3 Compare percents.** $90\% > 70\%$, so **Ben did better**, even with fewer points.

I SAY

"21 out of 30 is 70%, and 18 out of 20 is 90%."
"The bigger count does not decide it, because the wholes are different."

I WRITE

The two wholes are different, so **I compare percents: $90\% > 70\%$, so Ben did better.**

2 • DRAW IT*your turn: label the whole, the part and the percent*

Kira made 36 of 60 free throws. Dev made 32 of 40. Kira made more. Shade each bar to show each percent. Who is the better shooter?

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Kira: $36/60 = \underline{\hspace{2cm}}/10 = \underline{\hspace{2cm}}\%$.
- 2** Dev: $32/40 = \underline{\hspace{2cm}}/10 = \underline{\hspace{2cm}}\%$.
- 3** Shade each bar, then compare the percents.

3 • SAY IT*tell a partner*

Partner A: "_____ out of _____ is _____%, and _____ out of _____ is _____%."

Partner B: "So who did better? Why doesn't the bigger count decide it?"

4 • WRITE IT*pick a starter · name the whole · use units***WORD BANK** **out of** · **the whole** · **different wholes** · **compare** · **greater than (>)** · **better****Because** I know _____ is the better shooter because _____% > _____%.**But** Some students think _____ is better because _____, but _____.**So** The two wholes are different, so _____.
